

# VIP Validation Report

**Deployment:** CI Connect 2026.09.0 + WB 2026.09.0+174.pro3 + PM 2026.09.0  
**Generated:** 2026-09-19 20:44:01 UTC

## Products Under Test

| Product         | URL                   | Version            | Results                       |
|-----------------|-----------------------|--------------------|-------------------------------|
| Connect         | http://localhost:3939 | 2026.09.0          | 7 passed, 2 skipped (9 tests) |
| Workbench       | http://localhost:8787 | 2026.09.0+174.pro3 | 1 passed, 1 failed (2 tests)  |
| Package Manager | http://localhost:4242 | 2026.09.0          | 3 passed, 2 skipped (5 tests) |

## Summary

|         |      |
|---------|------|
| Total   | 25   |
| Passed  | 19   |
| Failed  | 2    |
| Skipped | 4    |
| Status  | FAIL |

## Provenance

|              |                                                   |
|--------------|---------------------------------------------------|
| VIP version  | 2026.9.2                                          |
| Run duration | 62.5s                                             |
| Python       | 3.14.7                                            |
| Platform     | Linux-6.17.0-1022-azure-x86_64-with-glibc2.39     |
| Mode         | full                                              |
| Exit status  | 1 (tests were collected and run, but some failed) |

## Failures & Skips

Every check that did not pass, in full. Passing checks are counted above, and every check appears in Detailed Results below.

### Failed (2)

**FAIL**

User can log in to Workbench via the web UI

Workbench

src/vip\_tests/workbench/test\_auth.py::test\_workbench\_login[chromium] 4.88s

Test procedure

- > Given Workbench is accessible at the configured URL
- > When a user navigates to the Workbench login page and enters valid credentials
- > Then the Workbench homepage is displayed
- > And the current user element is visible and non-empty in the header

test\_workbench\_login[chromium]: Login failed: Error: Temporary server error, please try again

```
fixturefunc = <function navigate_and_login at 0x7fcf26632fb0>
request = <FixtureRequest for <Function test_workbench_login[chromium]>>
kwargs = {'page': <Page url='http://localhost:8787/auth-sign-in?appUri=%2F&error=2'>,
          'workbench_url': 'http://localhost:8787', 'test_username': 'testuser', 'test_password':
          'bvyBqTCq5YXWHW4h', ...}
```

```

def call_fixture_func(
    fixturefunc: _FixtureFunc[FixtureValue], request: FixtureRequest, kwargs
) -> FixtureValue:
    if inspect.isgeneratorfunction(fixturefunc):
        fixturefunc = cast(Callable[..., Generator[FixtureValue]], fixturefunc)
        generator = fixturefunc(**kwargs)
        try:
            fixture_result = next(generator)
        except StopIteration:
            raise ValueError(f"{request.fixturename} did not yield a value") from None
        finalizer = functools.partial(_teardown_yield_fixture, fixturefunc, generator)
        request.addfinalizer(finalizer)
    else:
        fixturefunc = cast(Callable[..., FixtureValue], fixturefunc)
        fixture_result = fixturefunc(**kwargs)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

.venv/lib/python3.14/site-packages/_pytest/fixtures.py:1005:
src/vip_tests/workbench/test_auth.py:114: in navigate_and_login
    workbench_login(
-----

page = <Page url='http://localhost:8787/auth-sign-in?appUri=%2F&error=2'>
workbench_url = 'http://localhost:8787', username = 'testuser'
password = 'bvYBqTCq5YXWHW4h', auth_provider = 'password'
interactive_auth = False, auth_mode = 'none', workbench_auth_error = None
max_retries = 3, retry_delay = 2.0

def workbench_login(
    page: Page,
    workbench_url: str,
    username: str,
    password: str,
    auth_provider: str = "password",
    interactive_auth: bool = False,
    *,
    auth_mode: str = "none",
    workbench_auth_error: str | None = None,
    max_retries: int = 3,
    retry_delay: float = 2.0,
) -> None:
    """Navigate to Workbench homepage, logging in only if required.

    This function:
    - Navigates directly to Workbench's URL
    - Handles OIDC/SSO via pre-loaded storage state (--interactive-auth / --headless-auth)
    - Only fills login form for password auth
    - Retries on transient server errors (e.g., too many logins)

    Args:
    page: Playwright page object
    workbench_url: Base URL for Workbench (e.g., http://localhost:8787)
    username: Login username
    password: Login password
    auth_provider: Auth type (e.g., "password", "oidc", "saml")
    interactive_auth: True when an auth session is pre-loaded (either
        --interactive-auth or --headless-auth)
    auth_mode: Active auth mode ("interactive", "headless", or "none"),
        used to name the responsible CLI flag in skip messages
    workbench_auth_error: Reason the pre-test auth flow could not
        establish a Workbench session, if known. Quoted in the skip
        message so users see the real cause instead of a guess.
    max_retries: Max login attempts on transient errors (default 3)
    retry_delay: Seconds to wait between retries (default 2.0)

    Raises:
    pytest.skip: For non-password auth without a pre-loaded auth session,
        or when the session's storage state doesn't cover Workbench

```

```

        AssertionError: When password login fails after retries

"""
homepage_logo = page.locator(Hompage.POSIT_LOGO)

# For non-password auth without a pre-loaded auth session, skip immediately
if auth_provider != "password" and not interactive_auth:
    pytest.skip(
        f"Login form not available for auth provider {auth_provider!r}. "
        "Pass --interactive-auth or --headless-auth to pre-load browser storage state."
    )

page.goto(workbench_url)
page.wait_for_load_state("load")

# Fast path: already logged in (common with interactive_auth)?
if homepage_logo.is_visible():
    return

# A valid session cookie can redirect straight into a running session's IDE
# view instead of the homepage -- that view has none of Homepage's chrome, so
# the check above misses it and the login-page probe below also misses it
# (it's neither a login page nor the homepage). Same case test_sessions.py
# handles when navigating back from a session: go to /home explicitly.
if "/s/" in page.url:
    page.goto(f"{workbench_url}/home")
    page.wait_for_load_state("load")
    if homepage_logo.is_visible():
        return

# Check if we landed on a login/IdP page
if _on_login_page(page.url):
    # The sign-in page renders client-side after ``load``; wait once for
    # either the password form's username field or an OIDC "Sign in with ..."
    # button to appear before deciding which flow applies. A short fixed wait
    # on only the SSO button races the render and can misread a slow OIDC
    # sign-in page (e.g. an ``?error=2`` bounce) as a password deployment --
    # which then fails the retry loop with "Login failed after 3 attempts"
    # instead of skipping. Waiting for either control settles that race
    # without penalising real password deployments (the username field
    # appears promptly there).
    try:
        page.locator(f"{LoginPage.USERNAME}", button:has-text('Sign
            in')).first.wait_for(
            state="visible", timeout=TIMEOUT_PAGE_LOAD
        )
    except Exception: # noqa: BLE001
        pass

    # An OIDC sign-in page shows a "Sign in with ..." button and no username
    # field. The "sign in" role-name also matches a password form's submit
    # button, so the *absence* of the username field is what distinguishes a
    # true SSO-only page from a password form.
    sso_button = page.get_by_role("button", name=re.compile(r"sign in",
        re.IGNORECASE)).first
    sso_button_present = sso_button.is_visible()
    # A deployment can also skip its own sign-in page entirely and redirect
    # straight to the IdP, whose markup matches neither probe above (see
    # _external_idp_host). Landing off the Workbench origin is conclusive on
    # its own: no Workbench password form is reachable from there.
    idp_host = _external_idp_host(page.url, workbench_url)
    sso_only = idp_host is not None or (
        sso_button_present and not page.locator(LoginPage.USERNAME).is_visible()
    )

    if interactive_auth and sso_only:
        # Storage state was pre-loaded by --interactive-auth / --headless-auth.
        # Workbench's SSO sign-in page does not auto-redirect to the IdP; it renders a
        # "Sign in with OpenID" button. Clicking it triggers a silent SSO round-trip

```

```

# using the saved IdP cookies. The round-trip is serialized across xdist
# workers (see _silent_sso_signin / oidc_login_lock) to avoid storming the
# shared IdP session (#484/#467). When the deployment redirected us
# off-origin instead there is no such button to click, so go straight
# to the skip -- clicking a locator that resolves to nothing would
# raise a Playwright error in place of an actionable skip.
if sso_button_present and _silent_sso_signin(sso_button, homepage_logo,
workbench_url):
    return # Silent SSO succeeded
# No usable IdP session (expired, or storage state was stripped for the
# password-login test) – skip gracefully. Recompute the IdP host: the
# click above can navigate off-origin before timing out, so where we
# ended up is only knowable now, not before the attempt.
_skip_workbench_session_unproven(
    auth_mode=auth_mode,
    workbench_auth_error=workbench_auth_error,
    landed_url=page.url,
    idp_host=_external_idp_host(page.url, workbench_url),
)

if auth_provider != "password":
    _skip_workbench_session_unproven(
        auth_mode=auth_mode,
        workbench_auth_error=workbench_auth_error,
        landed_url=page.url,
        idp_host=_external_idp_host(page.url, workbench_url),
    )
# Even when auth_provider is reported as "password", the deployment may
# actually present an SSO/OIDC sign-in page (a "Sign in with ..." button
# and no username field) – e.g. auth_provider defaulted to "password" on
# a config-less run against an OIDC deployment. The password login form
# is unavailable there, so skip rather than fail the retry loop.
if sso_only:
    where = (
        f"redirects sign-in to the identity provider at {idp_host}"
        if idp_host
        else "presents an SSO/OIDC sign-in page instead (no username/password
            fields)"
    )
    pytest.skip(
        "This scenario requires a Workbench password-login form, but the "
        f"deployment {where}. Password login cannot be exercised on an SSO "
        "deployment, so this scenario is skipped."
    )
# Password auth – proceed with form login below
else:
    # Not on homepage, not on login page – unexpected state
    # Give it one more check in case page is still loading
    try:
        homepage_logo.wait_for(state="visible", timeout=TIMEOUT_QUICK)
        return
    except Exception: # noqa: BLE001
        pass

# Password authentication with retry logic
login_form = page.locator(LoginPage.USERNAME)
error_panel = page.locator(LoginPage.ERROR_PANEL)

for attempt in range(max_retries):
    if attempt > 0:
        time.sleep(retry_delay)
        page.goto(workbench_url)

    # Fast path check on retry
    if homepage_logo.is_visible():
        return

    # Wait for login form to be ready
    try:

```

```

        login_form.wait_for(state="visible", timeout=TIMEOUT_QUICK)
    except Exception: # noqa: BLE001
        continue

    # Fill and submit
    page.fill(LoginPage.USERNAME, username)
    page.fill(LoginPage.PASSWORD, password)

    stay_signed_in = page.locator(LoginPage.STAY_SIGNED_IN)
    if stay_signed_in.is_visible() and not stay_signed_in.is_checked():
        stay_signed_in.click()

    page.click(LoginPage.BUTTON)

    # Wait for either homepage (success) or error panel (failure)
    homepage_or_error = homepage_logo.or_(error_panel)
    try:
        homepage_or_error.wait_for(state="visible", timeout=TIMEOUT_PAGE_LOAD)
    except Exception as exc:
        if attempt == max_retries - 1:
            raise AssertionError(
                f"Login failed after {max_retries} attempts: no response"
            ) from exc
        continue

    # Check which one appeared
    if homepage_logo.is_visible():
        return # Success!

    # Error appeared - extract message and maybe retry
    if attempt == max_retries - 1:
        error_text = page.locator(LoginPage.ERROR_TEXT).text_content()
        raise AssertionError(f"Login failed: {error_text or 'Unknown error'}")
>
E       AssertionError: Login failed: Error: Temporary server error, please try again

src/vip_tests/workbench/conftest.py:1060: AssertionError

```

#### Likely causes:

- Incorrect test credentials in vip.toml or VIP\_TEST\_USERNAME/VIP\_TEST\_PASSWORD
- Auth provider misconfiguration (SAML, OIDC, LDAP, PAM)
- Workbench login page markup changed or is unreachable

#### Suggested next steps:

1. Verify credentials work by logging in manually at the Workbench URL
2. Check the auth provider configuration in rserver.conf / auth-pam or auth-saml settings
3. If using an external IdP, try `--interactive-auth` or `--headless-auth` instead of a password test

**Documentation:** [https://docs.posit.co/ide/server-pro/authenticating\\_users/authenticating\\_users.html](https://docs.posit.co/ide/server-pro/authenticating_users/authenticating_users.html)

#### **FAIL** This check intentionally fails to demonstrate failure rendering Prerequisites

*As someone evaluating a VIP report*

src/vip\_tests/prerequisites/test\_expected\_failure.py::test\_intentional\_failure 0.00s

#### Test procedure

- › Given a check that is written to fail on purpose
- › When the check runs as part of this example report
- › Then it fails by design, not because anything is actually broken

**test\_intentional\_failure: This failure is intentional: it exists only to show how a failed check renders in this example report. No product configuration will resolve it.**

**assert 'fail (by design)' == 'pass'**

**- pass**

**+ fail (by design)**

```

fixturefunc = <function check_fails_by_design at 0x7fcf267ce1f0>
request = <FixtureRequest for <Function test_intentional_failure>>
kwargs = {'demo_outcome': {'expected': 'pass', 'actual': 'fail (by design)'}}

```

```

def call_fixture_func(
    fixturefunc: _FixtureFunc[FixtureValue], request: FixtureRequest, kwargs
) -> FixtureValue:
    if inspect.isgeneratorfunction(fixturefunc):
        fixturefunc = cast(Callable[..., Generator[FixtureValue]], fixturefunc)
        generator = fixturefunc(**kwargs)
        try:
            fixture_result = next(generator)
        except StopIteration:
            raise ValueError(f"{request.fixturename} did not yield a value") from None
        finalizer = functools.partial(_teardown_yield_fixture, fixturefunc, generator)
        request.addfinalizer(finalizer)
    else:
        fixturefunc = cast(Callable[..., FixtureValue], fixturefunc)
    > fixture_result = fixturefunc(**kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

.venv/lib/python3.14/site-packages/_pytest/fixtures.py:1005:
-----
demo_outcome = {'expected': 'pass', 'actual': 'fail (by design)'}

    @then("it fails by design, not because anything is actually broken")
    def check_fails_by_design(demo_outcome):
    >     assert demo_outcome["actual"] == demo_outcome["expected"], (
        "This failure is intentional: it exists only to show how a failed check "
        "renders in this example report. No product configuration will resolve it."
    )
E       AssertionError: This failure is intentional: it exists only to show how a failed check
renders in this example report. No product configuration will resolve it.
E       assert 'fail (by design)' == 'pass'
E
E       - pass
E       + fail (by design)

src/vip_tests/prerequisites/test_expected_failure.py:49: AssertionError

```

#### Likely causes:

- This check is designed to always fail, so the example report has a failure card to show
- No product or configuration problem is involved

#### Suggested next steps:

1. No action needed. This failure is intentional and does not indicate a real problem
2. It only runs when VIP\_ENABLE\_EXPECTED\_FAILURE\_DEMO=1 is set

**Documentation:** <https://github.com/posit-dev/vip/issues/73>

## Skipped (4)

### SKIP Expected R versions are available on Connect Connect

No expected R versions specified in vip.toml [runtimes]

src/vip\_tests/connect/test\_runtime\_versions.py::test\_r\_versions 0.18s

#### Test procedure

- > Given Connect is accessible at the configured URL
- > And expected R versions are specified in vip.toml
- > When I query Connect for available R versions
- > Then all expected R versions are present

### SKIP Expected Python versions are available on Connect Connect

No expected Python versions specified in vip.toml [runtimes]

src/vip\_tests/connect/test\_runtime\_versions.py::test\_python\_versions 0.18s

#### Test procedure

- > Given Connect is accessible at the configured URL
- > And expected Python versions are specified in vip.toml
- > When I query Connect for available Python versions
- > Then all expected Python versions are present

**SKIP Bioconductor mirror is accessible** **Package Manager**

No Bioconductor repository configured in Package Manager

src/vip\_tests/package\_manager/test\_repos.py::test\_bioconductor\_mirror 0.04s

**Test procedure**

- › Given Package Manager is running
- › When I query the Bioconductor repository for the "BiocGenerics" package
- › Then the package is found in the repository

**SKIP OpenVSX mirror is accessible** **Package Manager**

No OpenVSX repository configured in Package Manager

src/vip\_tests/package\_manager/test\_repos.py::test\_opensvx\_mirror 0.04s

**Test procedure**

- › Given Package Manager is running
- › When I query the OpenVSX repository for the "golang.Go" extension
- › Then the package is found in the repository

## Detailed Results

### Connect

9 tests — 7 passed, 2 skipped

**PASS** User can log in via the web UI **Connect**

src/vip\_tests/connect/test\_auth.py::test\_connect\_login\_ui 1.37s

**Test procedure**

- › Given Connect is accessible at the configured URL
- › When a user navigates to the Connect login page
- › And enters valid credentials
- › Then the user is successfully authenticated
- › And the Connect dashboard is displayed

**PASS** API key authentication works **Connect**

src/vip\_tests/connect/test\_auth.py::test\_connect\_login\_api 0.36s

**Test procedure**

- › Given Connect is accessible at the configured URL
- › And a valid API key is configured
- › When I request the current user via the API
- › Then the API returns user information

**PASS** Admin user exists and has admin privileges **Connect**

src/vip\_tests/connect/test\_users.py::test\_admin\_user 0.36s

**Test procedure**

- › Given Connect is accessible at the configured URL
- › When I retrieve the current user profile
- › Then the user has admin privileges

**PASS** Users can be listed **Connect**

src/vip\_tests/connect/test\_users.py::test\_list\_users 0.35s

**Test procedure**

- › Given Connect is accessible at the configured URL
- › When I list all users
- › Then the user list is not empty
- › And the test user exists in the user list

**PASS** Groups can be listed **Connect**

src/vip\_tests/connect/test\_users.py::test\_list\_groups 0.36s

**Test procedure**

- › Given Connect is accessible at the configured URL
- › When I list all groups
- › Then the response is successful

**SKIP** Expected R versions are available on Connect **Connect**

No expected R versions specified in vip.toml [runtimes]

src/vip\_tests/connect/test\_runtime\_versions.py::test\_r\_versions 0.18s

**Test procedure**

- › Given Connect is accessible at the configured URL
- › And expected R versions are specified in vip.toml
- › When I query Connect for available R versions
- › Then all expected R versions are present

**SKIP** Expected Python versions are available on Connect **Connect**

No expected Python versions specified in vip.toml [runtimes]

src/vip\_tests/connect/test\_runtime\_versions.py::test\_python\_versions 0.18s

**Test procedure**

- › Given Connect is accessible at the configured URL
- › And expected Python versions are specified in vip.toml
- › When I query Connect for available Python versions
- › Then all expected Python versions are present

**PASS** Expected Quarto versions are available **Connect**

```
src/vip_tests/connect/test_runtime_versions.py::test_quarto_versions 0.35s
```

#### Test procedure

- › Given Connect is accessible at the configured URL
- › When I query Connect for available Quarto versions
- › Then at least one Quarto version is available

**PASS** Connect system checks can be run and the report downloaded **Connect**

```
src/vip_tests/connect/test_system_checks.py::test_connect_system_checks 13.42s
```

#### Test procedure

- › Given Connect is accessible at the configured URL
- › And a valid API key is configured
- › When I trigger a new system check run via the Connect API
- › Then the system check report is returned
- › And I can download the system check report artifact

## Package Manager

5 tests — 3 passed, 2 skipped

**PASS** CRAN mirror is accessible **Package Manager**

```
src/vip_tests/package_manager/test_repos.py::test_cran_mirror 0.20s
```

#### Test procedure

- › Given Package Manager is running
- › When I query the CRAN repository for the "Matrix" package
- › Then the package is found in the repository

**PASS** PyPI mirror is accessible **Package Manager**

```
src/vip_tests/package_manager/test_repos.py::test_pypi_mirror 0.08s
```

#### Test procedure

- › Given Package Manager is running
- › When I query the PyPI repository for the "requests" package
- › Then the package is found in the repository

**SKIP** Bioconductor mirror is accessible **Package Manager**

No Bioconductor repository configured in Package Manager

```
src/vip_tests/package_manager/test_repos.py::test_bioconductor_mirror 0.04s
```

#### Test procedure

- › Given Package Manager is running
- › When I query the Bioconductor repository for the "BiocGenerics" package
- › Then the package is found in the repository

**SKIP** OpenVSX mirror is accessible **Package Manager**

No OpenVSX repository configured in Package Manager

```
src/vip_tests/package_manager/test_repos.py::test_opensvx_mirror 0.04s
```

#### Test procedure

- › Given Package Manager is running
- › When I query the OpenVSX repository for the "golang.Go" extension
- › Then the package is found in the repository

**PASS** At least one repository is configured **Package Manager**

```
src/vip_tests/package_manager/test_repos.py::test_repo_exists 0.05s
```

#### Test procedure

- › Given Package Manager is running
- › When I list all repositories
- › Then at least one repository exists

## Prerequisites

6 tests — 5 passed, 1 failed

### PASS Connect server is reachable Prerequisites

src/vip\_tests/prerequisites/test\_components.py::test\_product\_server\_is\_reachable[Connect] 0.03s

#### Test procedure

- › Given <product> is configured in vip.toml
- › When I request the <product> health endpoint
- › Then the server responds with a successful status code

### PASS Package Manager server is reachable Prerequisites

src/vip\_tests/prerequisites/test\_components.py::test\_product\_server\_is\_reachable[Package Manager] 0.07s

#### Test procedure

- › Given <product> is configured in vip.toml
- › When I request the <product> health endpoint
- › Then the server responds with a successful status code

### FAIL This check intentionally fails to demonstrate failure rendering Prerequisites

As someone evaluating a VIP report

src/vip\_tests/prerequisites/test\_expected\_failure.py::test\_intentional\_failure 0.00s

#### Test procedure

- › Given a check that is written to fail on purpose
- › When the check runs as part of this example report
- › Then it fails by design, not because anything is actually broken

**test\_intentional\_failure: This failure is intentional: it exists only to show how a failed check renders in this example report. No product configuration will resolve it.**

**assert 'fail (by design)' == 'pass'**

**- pass**

**+ fail (by design)**

```
fixturefunc = <function check_fails_by_design at 0x7fcf267celf0>
request = <FixtureRequest for <Function test_intentional_failure>>
kwargs = {'demo_outcome': {'expected': 'pass', 'actual': 'fail (by design)'}}
```

```
def call_fixture_func(
    fixturefunc: _FixtureFunc[FixtureValue], request: FixtureRequest, kwargs
) -> FixtureValue:
    if inspect.isgeneratorfunction(fixturefunc):
        fixturefunc = cast(Callable[..., Generator[FixtureValue]], fixturefunc)
        generator = fixturefunc(**kwargs)
        try:
            fixture_result = next(generator)
        except StopIteration:
            raise ValueError(f"{request.fixturename} did not yield a value") from None
        finalizer = functools.partial(_teardown_yield_fixture, fixturefunc, generator)
        request.addfinalizer(finalizer)
    else:
        fixturefunc = cast(Callable[..., FixtureValue], fixturefunc)
        > fixture_result = fixturefunc(**kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
.venv/lib/python3.14/site-packages/_pytest/fixtures.py:1005:
- - - - -
demo_outcome = {'expected': 'pass', 'actual': 'fail (by design)'}
```

```
@then("it fails by design, not because anything is actually broken")
def check_fails_by_design(demo_outcome):
>     assert demo_outcome["actual"] == demo_outcome["expected"], (
           "This failure is intentional: it exists only to show how a failed check "
           "renders in this example report. No product configuration will resolve it."
       )
E       AssertionError: This failure is intentional: it exists only to show how a failed check
renders in this example report. No product configuration will resolve it.
E       assert 'fail (by design)' == 'pass'
E
```

```
E      - pass
E      + fail (by design)
```

```
src/vip_tests/prerequisites/test_expected_failure.py:49: AssertionError
```

**Likely causes:**

- This check is designed to always fail, so the example report has a failure card to show
- No product or configuration problem is involved

**Suggested next steps:**

1. No action needed. This failure is intentional and does not indicate a real problem
2. It only runs when VIP\_ENABLE\_EXPECTED\_FAILURE\_DEMO=1 is set

**Documentation:** <https://github.com/posit-dev/vip/issues/73>

**PASS** Connect version matches configuration **Prerequisites**

```
src/vip_tests/prerequisites/test_versions.py::test_connect_version 0.18s
```

**Test procedure**

- › Given Connect is configured in vip.toml with a version expectation
- › When I fetch the Connect server version
- › Then the Connect version matches the configured value

**PASS** Package Manager version matches configuration **Prerequisites**

```
src/vip_tests/prerequisites/test_versions.py::test_package_manager_version 0.09s
```

**Test procedure**

- › Given Package Manager is configured in vip.toml with a version expectation
- › When I fetch the Package Manager server version
- › Then the Package Manager version matches the configured value

**PASS** Workbench server is reachable **Prerequisites**

```
src/vip_tests/prerequisites/test_components.py::test_product_server_is_reachable[Workbench]
0.11s
```

**Test procedure**

- › Given <product> is configured in vip.toml
- › When I request the <product> health endpoint
- › Then the server responds with a successful status code

## Security

3 tests — 3 passed

**PASS** Unauthenticated API request returns 401 **Security**

```
src/vip_tests/security/test_error_handling.py::test_unauthenticated_api_request_returns_401
0.01s
```

**Test procedure**

- › Given Connect is configured in vip.toml
- › When I make an unauthenticated API request to Connect
- › Then the response status is 401

**PASS** Invalid API key returns 401 **Security**

```
src/vip_tests/security/test_error_handling.py::test_invalid_api_key_returns_401 2.73s
```

**Test procedure**

- › Given Connect is configured in vip.toml
- › When I make an API request to Connect with an invalid key
- › Then the response status is 401

**PASS** Non-existent endpoint returns 404 **Security**

```
src/vip_tests/security/test_error_handling.py::test_nonexistent_endpoint_returns_404 0.01s
```

**Test procedure**

- › Given Connect is configured in vip.toml
- › When I request a non-existent endpoint on Connect
- › Then the response status is 404

## Workbench

2 tests — 1 passed, 1 failed

### PASS User can sign out of Workbench Workbench

src/vip\_tests/workbench/test\_auth.py::test\_workbench\_signout[chromium] 0.63s

#### Test procedure

- › Given Workbench is accessible and I am logged in
- › When I sign out of Workbench
- › Then I am redirected to the Workbench login page

### FAIL User can log in to Workbench via the web UI Workbench

src/vip\_tests/workbench/test\_auth.py::test\_workbench\_login[chromium] 4.88s

#### Test procedure

- › Given Workbench is accessible at the configured URL
- › When a user navigates to the Workbench login page and enters valid credentials
- › Then the Workbench homepage is displayed
- › And the current user element is visible and non-empty in the header

**test\_workbench\_login[chromium]: Login failed: Error: Temporary server error, please try again**

```
fixturefunc = <function navigate_and_login at 0x7fcf26632fb0>
request = <FixtureRequest for <Function test_workbench_login[chromium]>>
kwargs = {'page': <Page url='http://localhost:8787/auth-sign-in?appUri=%2F&error=2'>,
          'workbench_url': 'http://localhost:8787', 'test_username': 'testuser', 'test_password':
          'bvyBqTCq5YXWHW4h', ...}

def call_fixture_func(
    fixturefunc: _FixtureFunc[FixtureValue], request: FixtureRequest, kwargs
) -> FixtureValue:
    if inspect.isgeneratorfunction(fixturefunc):
        fixturefunc = cast(Callable[..., Generator[FixtureValue]], fixturefunc)
        generator = fixturefunc(**kwargs)
        try:
            fixture_result = next(generator)
        except StopIteration:
            raise ValueError(f"{request.fixturename} did not yield a value") from None
        finalizer = functools.partial(_teardown_yield_fixture, fixturefunc, generator)
        request.addfinalizer(finalizer)
    else:
        fixturefunc = cast(Callable[..., FixtureValue], fixturefunc)
        > fixture_result = fixturefunc(**kwargs)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

.venv/lib/python3.14/site-packages/_pytest/fixtures.py:1005:
-----
src/vip_tests/workbench/test_auth.py:114: in navigate_and_login
    workbench_login(
-----

page = <Page url='http://localhost:8787/auth-sign-in?appUri=%2F&error=2'>
workbench_url = 'http://localhost:8787', username = 'testuser'
password = 'bvyBqTCq5YXWHW4h', auth_provider = 'password'
interactive_auth = False, auth_mode = 'none', workbench_auth_error = None
max_retries = 3, retry_delay = 2.0

def workbench_login(
    page: Page,
    workbench_url: str,
    username: str,
    password: str,
    auth_provider: str = "password",
    interactive_auth: bool = False,
    *,
    auth_mode: str = "none",
    workbench_auth_error: str | None = None,
    max_retries: int = 3,
    retry_delay: float = 2.0,
) -> None:
    """Navigate to Workbench homepage, logging in only if required.
```

```

This function:
- Navigates directly to Workbench's URL
- Handles OIDC/SSO via pre-loaded storage state (--interactive-auth / --headless-auth)
- Only fills login form for password auth
- Retries on transient server errors (e.g., too many logins)

Args:
  page: Playwright page object
  workbench_url: Base URL for Workbench (e.g., http://localhost:8787)
  username: Login username
  password: Login password
  auth_provider: Auth type (e.g., "password", "oidc", "saml")
  interactive_auth: True when an auth session is pre-loaded (either
    --interactive-auth or --headless-auth)
  auth_mode: Active auth mode ("interactive", "headless", or "none"),
    used to name the responsible CLI flag in skip messages
  workbench_auth_error: Reason the pre-test auth flow could not
    establish a Workbench session, if known. Quoted in the skip
    message so users see the real cause instead of a guess.
  max_retries: Max login attempts on transient errors (default 3)
  retry_delay: Seconds to wait between retries (default 2.0)

Raises:
  pytest.skip: For non-password auth without a pre-loaded auth session,
    or when the session's storage state doesn't cover Workbench
  AssertionError: When password login fails after retries

"""
homepage_logo = page.locator(Hompage.POSIT_LOGO)

# For non-password auth without a pre-loaded auth session, skip immediately
if auth_provider != "password" and not interactive_auth:
    pytest.skip(
        f"Login form not available for auth provider {auth_provider!r}. "
        "Pass --interactive-auth or --headless-auth to pre-load browser storage state."
    )

page.goto(workbench_url)
page.wait_for_load_state("load")

# Fast path: already logged in (common with interactive_auth)?
if homepage_logo.is_visible():
    return

# A valid session cookie can redirect straight into a running session's IDE
# view instead of the homepage -- that view has none of Homepage's chrome, so
# the check above misses it and the login-page probe below also misses it
# (it's neither a login page nor the homepage). Same case test_sessions.py
# handles when navigating back from a session: go to /home explicitly.
if "/s/" in page.url:
    page.goto(f"{workbench_url}/home")
    page.wait_for_load_state("load")
    if homepage_logo.is_visible():
        return

# Check if we landed on a login/IdP page
if _on_login_page(page.url):
    # The sign-in page renders client-side after ``load``; wait once for
    # either the password form's username field or an OIDC "Sign in with ..."
    # button to appear before deciding which flow applies. A short fixed wait
    # on only the SSO button races the render and can misread a slow OIDC
    # sign-in page (e.g. an ``?error=2`` bounce) as a password deployment --
    # which then fails the retry loop with "Login failed after 3 attempts"
    # instead of skipping. Waiting for either control settles that race
    # without penalising real password deployments (the username field
    # appears promptly there).
    try:
        page.locator(f"{LoginPage.USERNAME}", button:has-text('Sign
            in')").first.wait_for(

```

```

        state="visible", timeout=TIMEOUT_PAGE_LOAD
    )
except Exception: # noqa: BLE001
    pass

# An OIDC sign-in page shows a "Sign in with ..." button and no username
# field. The "sign in" role-name also matches a password form's submit
# button, so the *absence* of the username field is what distinguishes a
# true SSO-only page from a password form.
sso_button = page.get_by_role("button", name=re.compile(r"sign in",
    re.IGNORECASE)).first
sso_button_present = sso_button.is_visible()
# A deployment can also skip its own sign-in page entirely and redirect
# straight to the IdP, whose markup matches neither probe above (see
# _external_idp_host). Landing off the Workbench origin is conclusive on
# its own: no Workbench password form is reachable from there.
idp_host = _external_idp_host(page.url, workbench_url)
sso_only = idp_host is not None or (
    sso_button_present and not page.locator(LoginPage.USERNAME).is_visible()
)

if interactive_auth and sso_only:
    # Storage state was pre-loaded by --interactive-auth / --headless-auth.
    # Workbench's SSO sign-in page does not auto-redirect to the IdP; it renders a
    # "Sign in with OpenID" button. Clicking it triggers a silent SSO round-trip
    # using the saved IdP cookies. The round-trip is serialized across xdist
    # workers (see _silent_sso_signin / oidc_login_lock) to avoid storming the
    # shared IdP session (#484/#467). When the deployment redirected us
    # off-origin instead there is no such button to click, so go straight
    # to the skip -- clicking a locator that resolves to nothing would
    # raise a Playwright error in place of an actionable skip.
    if sso_button_present and _silent_sso_signin(sso_button, homepage_logo,
        workbench_url):
        return # Silent SSO succeeded
    # No usable IdP session (expired, or storage state was stripped for the
    # password-login test) – skip gracefully. Recompute the IdP host: the
    # click above can navigate off-origin before timing out, so where we
    # ended up is only knowable now, not before the attempt.
    _skip_workbench_session_unproven(
        auth_mode=auth_mode,
        workbench_auth_error=workbench_auth_error,
        landed_url=page.url,
        idp_host=_external_idp_host(page.url, workbench_url),
    )

if auth_provider != "password":
    _skip_workbench_session_unproven(
        auth_mode=auth_mode,
        workbench_auth_error=workbench_auth_error,
        landed_url=page.url,
        idp_host=_external_idp_host(page.url, workbench_url),
    )

# Even when auth_provider is reported as "password", the deployment may
# actually present an SSO/OIDC sign-in page (a "Sign in with ..." button
# and no username field) – e.g. auth_provider defaulted to "password" on
# a config-less run against an OIDC deployment. The password login form
# is unavailable there, so skip rather than fail the retry loop.
if sso_only:
    where = (
        f"redirects sign-in to the identity provider at {idp_host}"
        if idp_host
        else "presents an SSO/OIDC sign-in page instead (no username/password
            fields)"
    )
    pytest.skip(
        f"This scenario requires a Workbench password-login form, but the "
        f"deployment {where}. Password login cannot be exercised on an SSO "
        f"deployment, so this scenario is skipped."
    )

```

```

    # Password auth - proceed with form login below
else:
    # Not on homepage, not on login page - unexpected state
    # Give it one more check in case page is still loading
    try:
        homepage_logo.wait_for(state="visible", timeout=TIMEOUT_QUICK)
        return
    except Exception: # noqa: BLE001
        pass

# Password authentication with retry logic
login_form = page.locator(LoginPage.USERNAME)
error_panel = page.locator(LoginPage.ERROR_PANEL)

for attempt in range(max_retries):
    if attempt > 0:
        time.sleep(retry_delay)
        page.goto(workbench_url)

    # Fast path check on retry
    if homepage_logo.is_visible():
        return

    # Wait for login form to be ready
    try:
        login_form.wait_for(state="visible", timeout=TIMEOUT_QUICK)
    except Exception: # noqa: BLE001
        continue

    # Fill and submit
    page.fill(LoginPage.USERNAME, username)
    page.fill(LoginPage.PASSWORD, password)

    stay_signed_in = page.locator(LoginPage.STAY_SIGNED_IN)
    if stay_signed_in.is_visible() and not stay_signed_in.is_checked():
        stay_signed_in.click()

    page.click(LoginPage.BUTTON)

    # Wait for either homepage (success) or error panel (failure)
    homepage_or_error = homepage_logo.or_(error_panel)
    try:
        homepage_or_error.wait_for(state="visible", timeout=TIMEOUT_PAGE_LOAD)
    except Exception as exc:
        if attempt == max_retries - 1:
            raise AssertionError(
                f"Login failed after {max_retries} attempts: no response"
            ) from exc
        continue

    # Check which one appeared
    if homepage_logo.is_visible():
        return # Success!

    # Error appeared - extract message and maybe retry
    if attempt == max_retries - 1:
        error_text = page.locator(LoginPage.ERROR_TEXT).text_content()
        raise AssertionError(f"Login failed: {error_text or 'Unknown error'}")
>
E       AssertionError: Login failed: Error: Temporary server error, please try again

src/vip_tests/workbench/confptest.py:1060: AssertionError

```

**Likely causes:**

- Incorrect test credentials in vip.toml or VIP\_TEST\_USERNAME/VIP\_TEST\_PASSWORD
- Auth provider misconfiguration (SAML, OIDC, LDAP, PAM)
- Workbench login page markup changed or is unreachable

**Suggested next steps:**

1. Verify credentials work by logging in manually at the Workbench URL
2. Check the auth provider configuration in rserver.conf / auth-pam or auth-saml settings
3. If using an external IdP, try `--interactive-auth` or `--headless-auth` instead of a password test

**Documentation:** [https://docs.posit.co/ide/server-pro/authenticating\\_users/authenticating\\_users.html](https://docs.posit.co/ide/server-pro/authenticating_users/authenticating_users.html)